

Jason Yang

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Education	Massachusetts Institute of Technology (MIT) Candidate for B.S. in Electrical Engineering with Computing	Cambridge, MA May 2029
Projects	Low-Power Desk Display - Designed and built a custom PCB integrating an ESP32-S3 and power regulation, interfacing with a Raspberry Pi and display to create a compact holographic-style desk display featuring weather, music playback, calendar tasks, and Bluetooth headphone status - Developed firmware, graphics rendering, and a custom communication protocol over SPI with dedicated acknowledgement signaling for reliable cross-device communication String Art Robot - Designed and built a robot as part of a team to recreate digital images using string threaded along a board of 200 nails - Developed C++ image processing algorithms and tooling to generate optimized stringing paths and control an Arduino-based stringing robot - Successfully drew multiple pieces, including the Mona Lisa and the MIT Dome	June 2026 - Present October 2025 - December 2025
Experience	MIT Space Propulsion Lab Undergraduate Researcher - Developing mission control and flight software for a 3U-XL CubeSat designed to validate staged electrospray propulsion technology for deep space exploration - Conducting end-to-end satellite integration and system-level verification to validate hardware, software, and subsystem interfaces for flight MIT Rocket Team Controls Lead - Implementing a closed-loop throttle control system for a bipropellant liquid engine, regulating oxidizer and fuel mass flow rates - Writing embedded drivers and flight firmware for a propeller-powered VTVL hopper - Developing a hardware-in-the-loop framework integrating flight hardware with simulated vehicle dynamics to validate sensor interfaces, state estimation, and control algorithms for a propulsive lander MIT Media Lab - Responsive Environments Undergraduate Researcher - Reverse-engineered firmware for a smart BLE wearable ring to enable real-time sensor data streaming and integrated TensorFlow for gesture recognition - Prototyping a dense modular capacitive/resistive sensor array for use on fabrics The RoboLancers Programming Lead, Project Manager - Led a team of 8-10 programmers, developing award-winning robots emphasizing maintainability and broad team participation - Created a programming curriculum, developing programming competency across team of 50+ members - Co-led a project management effort for 100+ members to streamline core workflows, reducing initial robot build time by over 10%	Cambridge, MA May 2026 - Present Cambridge, MA August 2025 - Present Philadelphia, PA June 2023 - June 2025
Skills	C, C++, Python, Java, JavaScript/TypeScript, STM32, Embedded Systems, Git, KiCad	